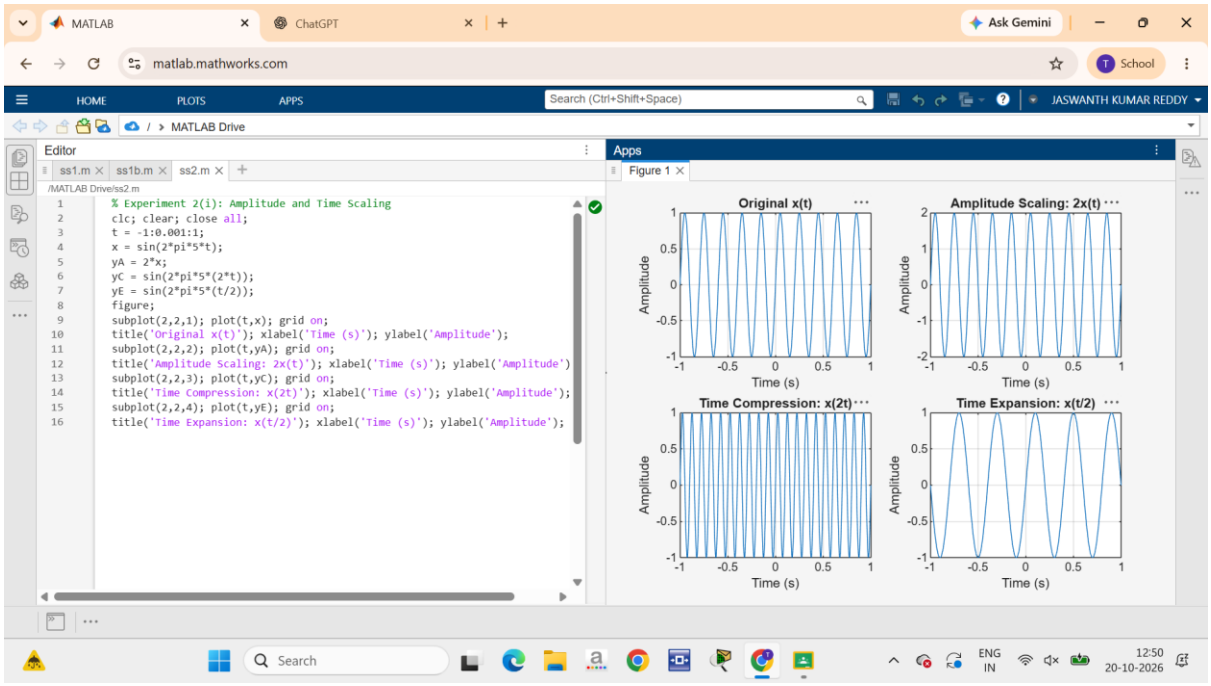
Experiment No. 2

Signal Operations and Transformations

OBJECTIVE:

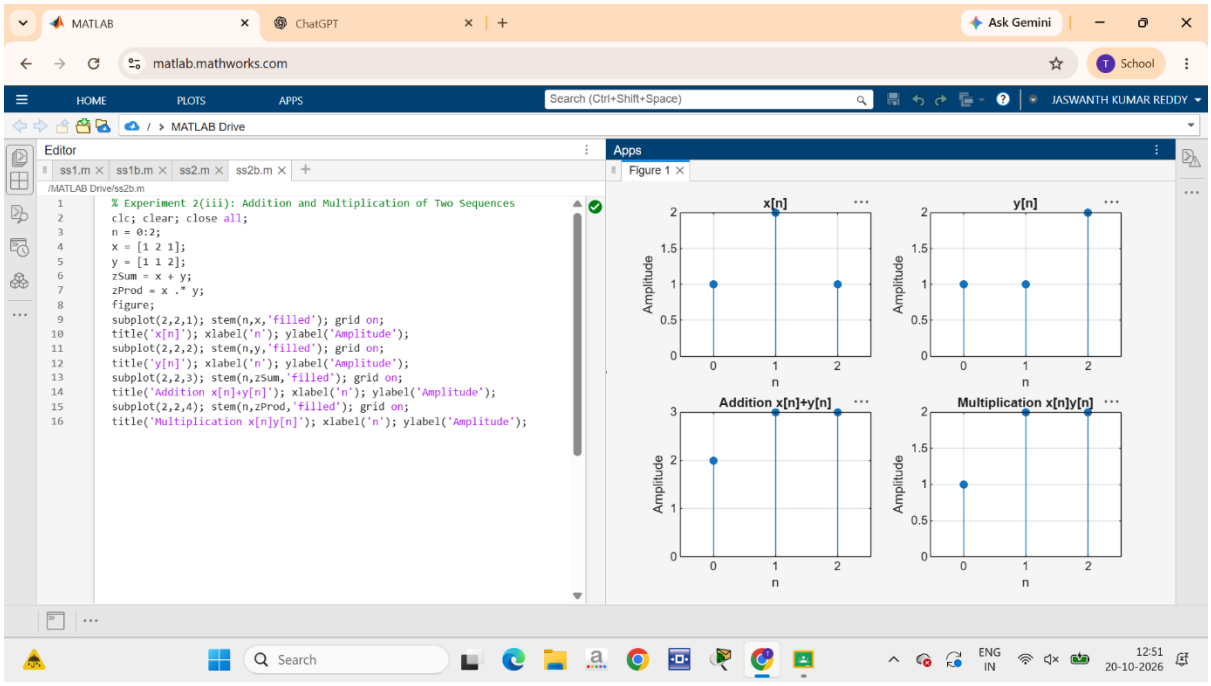
(i) To perform amplitude scaling and time scaling of continuous-time and discrete-time signals and analyze their effects on signal characteristics.

Output Graphs:



(iii) To perform addition and multiplication of two discrete-time sequences and analyze the resulting signal characteristics

Output Graphs:


MATLAB
ChatGPT
Ask Gemini
matlab.mathworks.com
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APPS
Search (Ctrl+Shift+Space)
JASWANTH KUMAR REDDY
MATLAB Drive
Editor
ss1.m
ss1b.m
ss2.m
ss2b.m
/MATLAB Drive/ss2b.m
% Experiment 2(iii): Addition and Multiplication of Two Sequences
clc; clear; close all;
n = 0:2;
x = [1 2 1];
y = [1 1 2];
zSum = x + y;
zProd = x .* y;
figure;
subplot(2,2,1); stem(n,x,'filled'); grid on;
title('x[n]'); xlabel('n'); ylabel('Amplitude');
subplot(2,2,2); stem(n,y,'filled'); grid on;
title('y[n]'); xlabel('n'); ylabel('Amplitude');
subplot(2,2,3); stem(n,zSum,'filled'); grid on;
title('Addition x[n]+y[n]'); xlabel('n'); ylabel('Amplitude');
subplot(2,2,4); stem(n,zProd,'filled'); grid on;
title('Multiplication x[n]y[n]'); xlabel('n'); ylabel('Amplitude');
Apps
Figure 1
x[n]
y[n]
Addition x[n]+y[n]
Multiplication x[n]y[n]
Amplitude
n
Search
ENG
IN
12:51
20-10-2026